Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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Contents

1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise**: how unexpected a contribution is given prior context
- **Transience**: how quickly its influence fades
- **Resonance**: novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,
- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
  - **Human / AI** in Human-AI
  - **Starter-side / Non-starter-side** in same-type conditions
- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper's narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as resonance $\sim$ surprise $\times$ slot $+ (1 \mid$ conversation_id$)$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) \;=\; \left|\,\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}\,\right|,$$

extracted via `emtrends`. Slot mean differences are retained as a supplemental view. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	39	351
human-ai	48	431
human-human	31	279

3 Preprocessing

```
## Slot-level rows: 2115
```

```
## Metric rows: 6345
```

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: |slope_1 - slope_2| from resonance ~ surprise * slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	0.978	0.856	-0.122	0.047	-2.607	0.009	0.122	**
Human-Human	0.880	0.913	0.033	0.054	0.614	0.540	0.033	
AI-AI	0.993	1.011	0.017	0.042	0.413	0.680	0.017	

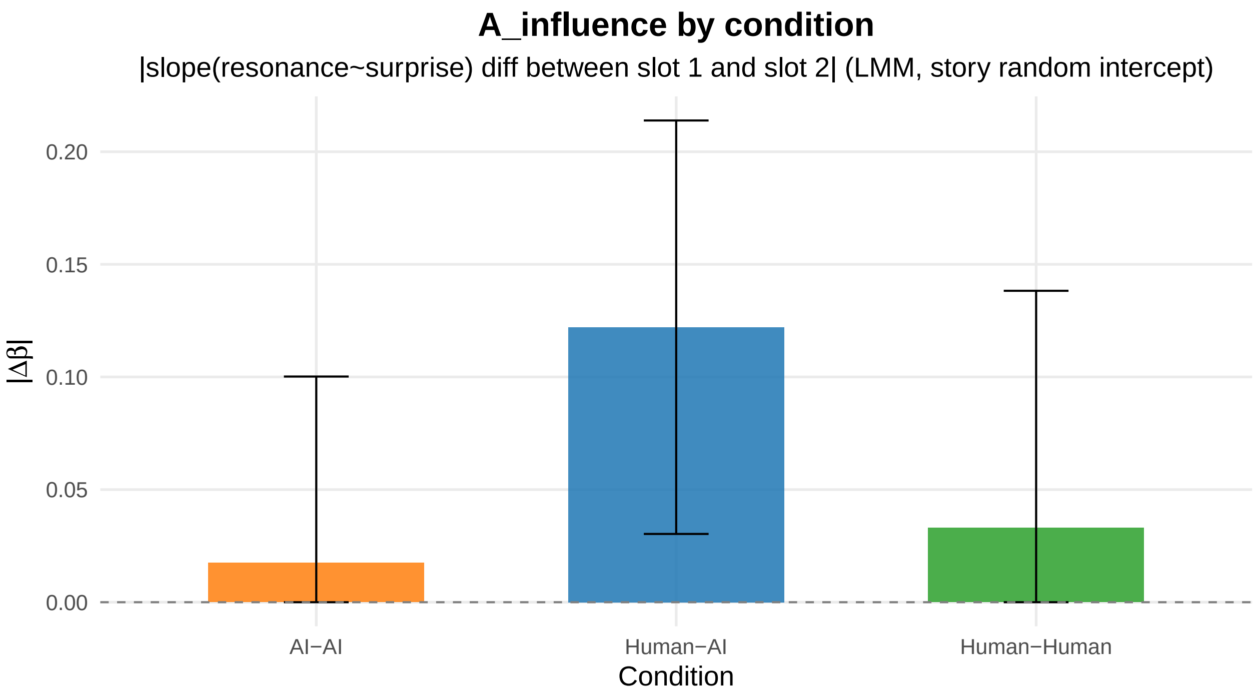


Table 3: Robustness: per-story |slope_1 - slope_2| (OLS per story, 4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	48	0.304	0.369
Human-Human	31	0.255	0.316
AI-AI	39	0.195	0.274

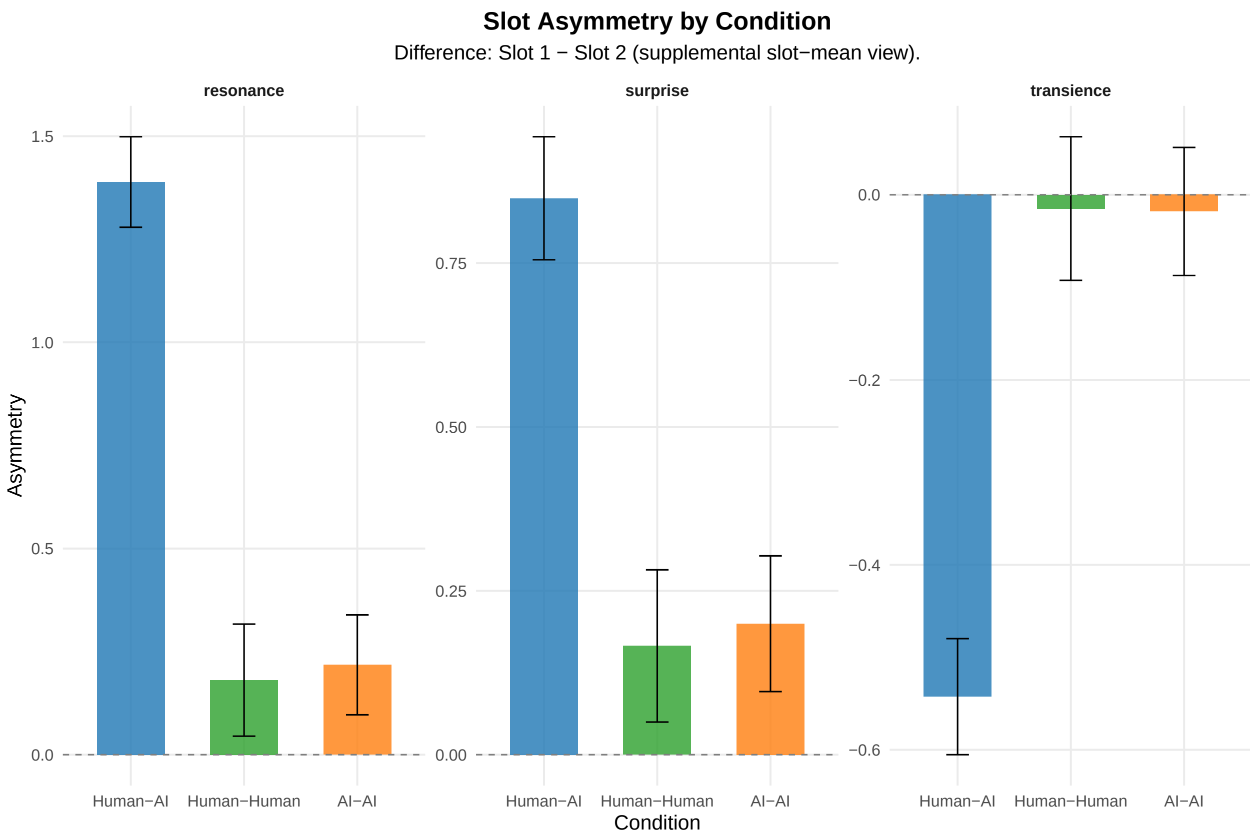
```
##
## Kruskal-Wallis on per-story A_influence:
##
##  Kruskal-Wallis rank sum test
##
## data:  A_influence_story by condition
## Kruskal-Wallis chi-squared = 3.4891, df = 2, p-value = 0.1747
```

5 Supplemental: Slot Mean Comparison

Table 4: Slot asymmetry by condition for novelty metrics

metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p	sig
surprise	Human-AI	-1.067	-1.916	0.849	0.048	17.731	0.000	***
surprise	Human-Human	-1.087	-1.253	0.166	0.059	2.800	0.005	**
surprise	AI-AI	-1.651	-1.851	0.200	0.053	3.782	0.000	***
transience	Human-AI	-0.940	-0.398	-0.543	0.032	-16.951	0.000	***
transience	Human-Human	-0.426	-0.412	-0.015	0.040	-0.376	0.707	
transience	AI-AI	-0.902	-0.884	-0.018	0.035	-0.512	0.609	

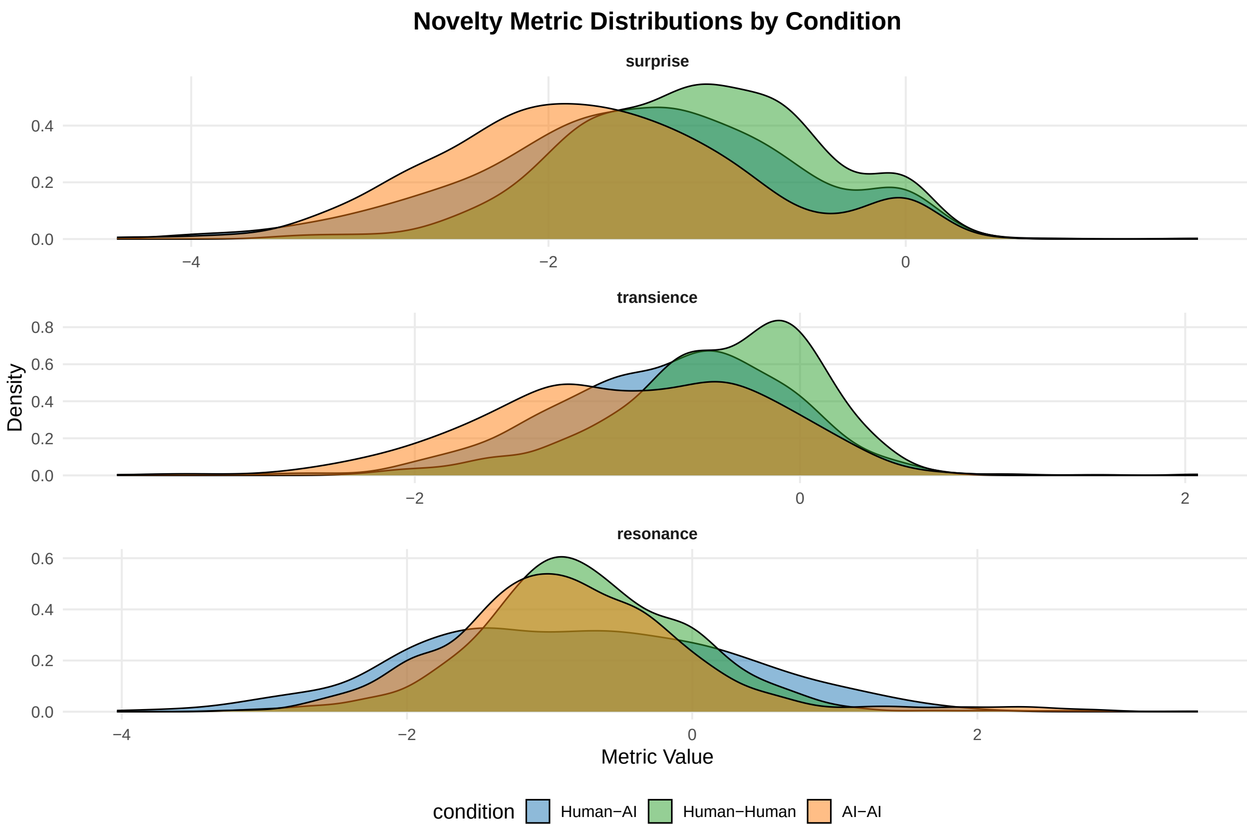
metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p	sig
resonance	Human-AI	-0.127	-1.516	1.389	0.056	24.796	0.000	***
resonance	Human-Human	-0.660	-0.841	0.181	0.069	2.607	0.009	**
resonance	AI-AI	-0.749	-0.967	0.218	0.062	3.524	0.000	***



6 Supplemental: General Condition-Level Comparison

Table 5: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	48	855	-1.488	0.866
Human-AI	transience	48	855	-0.673	0.602
Human-AI	resonance	48	855	-0.816	1.103
Human-Human	surprise	31	558	-1.170	0.685
Human-Human	transience	31	558	-0.419	0.532
Human-Human	resonance	31	558	-0.751	0.715
AI-AI	surprise	39	702	-1.751	0.844
AI-AI	transience	39	702	-0.893	0.700
AI-AI	resonance	39	702	-0.858	0.866




```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
##    Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4989.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2204 -0.6455 -0.0392  0.6002  4.2362
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.09418  0.3069
##  Residual                    0.57142  0.7559
## Number of obs: 2115, groups:  conversation_id, 118
##
## Fixed effects:
##                        Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)            -1.48795    0.05129 115.45950 -29.010  < 2e-16 ***
## conditionHuman-Human    0.31820    0.08181 115.07266   3.890 0.000169 ***
## conditionAI-AI         -0.26282    0.07655 115.10821  -3.433 0.000829 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##             (Intr) cndH-H
## cndtnHmn-Hm -0.627
## condtnAI-AI -0.670  0.420
##  contrast                   estimate     SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)   -0.318 0.0818 115  -3.890  0.0005
##  (Human-AI) - (AI-AI)          0.263 0.0765 115   3.433  0.0025
##  (Human-Human) - (AI-AI)       0.581 0.0854 115   6.805  <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
##    Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 3359.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.7827 -0.5717  0.0652  0.6475  4.3653
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.1361   0.3689
##  Residual                    0.2501   0.5001
## Number of obs: 2115, groups:  conversation_id, 118
##
## Fixed effects:
##                       Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)           -0.67134    0.05593 115.20725 -12.002  < 2e-16 ***
## conditionHuman-Human   0.25232    0.08926 115.06072   2.827  0.00555 **
## conditionAI-AI        -0.22181    0.08352 115.07419  -2.656  0.00903 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.627
## condtnAI-AI -0.670  0.420
##  contrast                   estimate     SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)   -0.252 0.0893 115  -2.827  0.0166
##  (Human-AI) - (AI-AI)          0.222 0.0835 115   2.656  0.0271
##  (Human-Human) - (AI-AI)       0.474 0.0932 115   5.087  <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
```

```
##    Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 5732.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4398 -0.6497 -0.0537  0.6140  4.6625
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.0000   0.0000
##  Residual                    0.8755   0.9357
## Number of obs: 2115, groups:  conversation_id, 118
##
## Fixed effects:
##                       Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)           -0.81562    0.03200 2112.00000 -25.489   <2e-16 ***
## conditionHuman-Human   0.06489    0.05092 2112.00000   1.274    0.203
## conditionAI-AI        -0.04199    0.04766 2112.00000  -0.881    0.378
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.628
## condtnAI-AI -0.671  0.422
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##  contrast                    estimate     SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  -0.0649 0.0509 115  -1.274  0.6152
##  (Human-AI) - (AI-AI)         0.0420 0.0477 115   0.881  1.0000
##  (Human-Human) - (AI-AI)      0.1069 0.0531 114   2.014  0.1390
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Pooled Novelty-to-Resonance Relationship

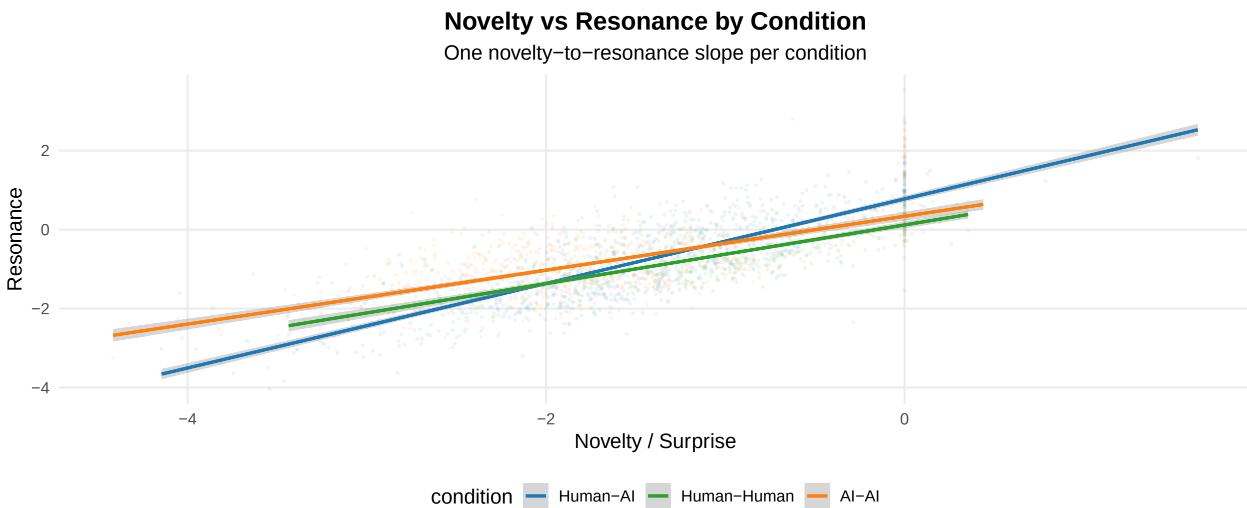
```
## Novelty-to-resonance model:
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
##    Data: df_slots
##
## REML criterion at convergence: 3334.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.6733 -0.6298 -0.0692  0.5579  5.5157
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
```

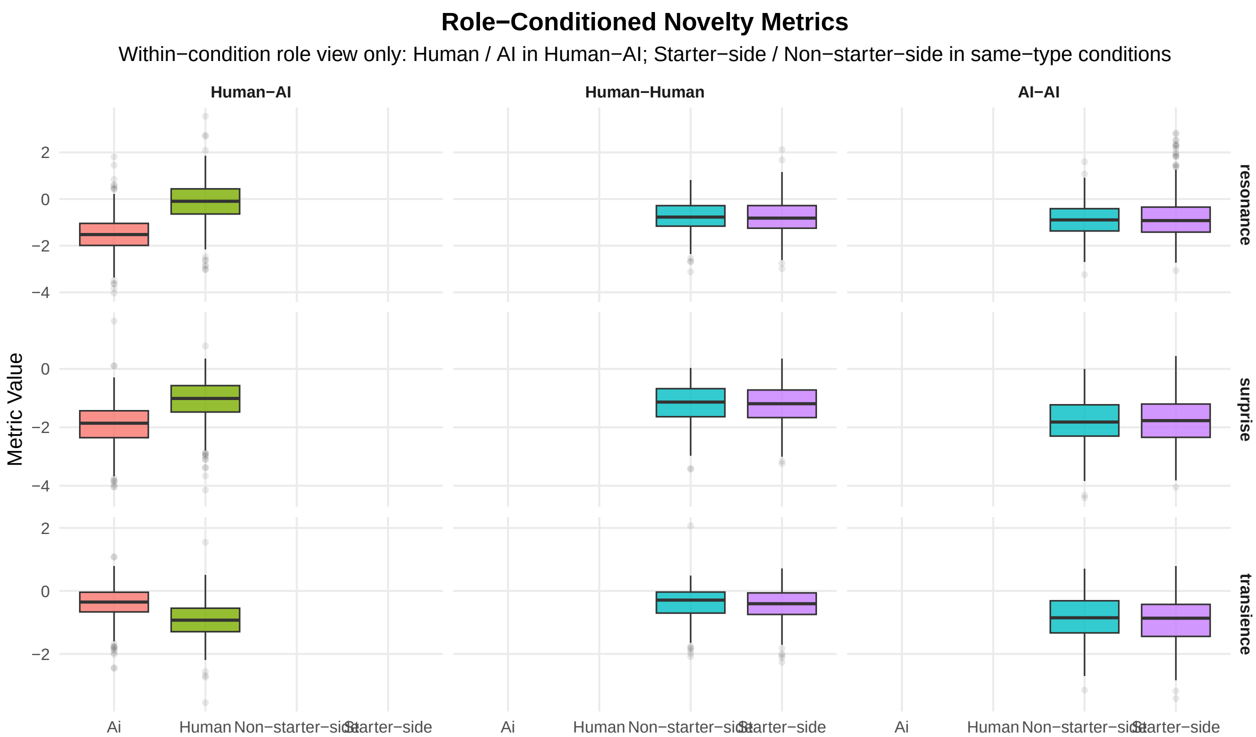
```
##  conversation_id (Intercept) 0.1314   0.3625
##  Residual                    0.2458   0.4958
## Number of obs: 2115, groups:  conversation_id, 118
##
## Fixed effects:
##                                Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                   8.454e-01  6.290e-02  1.849e+02  13.440  < 2e-16
## surprise                      1.117e+00  2.050e-02  2.027e+03  54.482  < 2e-16
## conditionHuman-Human         -5.433e-01  1.010e-01  1.885e+02  -5.376 2.23e-07
## conditionAI-AI                6.685e-03  9.864e-02  2.195e+02   0.068    0.946
## surprise:conditionHuman-Human -2.169e-01  3.963e-02  2.058e+03  -5.473 4.96e-08
## surprise:conditionAI-AI       -1.404e-01  3.301e-02  2.076e+03  -4.254 2.19e-05
##
## (Intercept)                   ***
## surprise                      ***
## conditionHuman-Human          ***
## conditionAI-AI
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI       ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise     0.485
## cndtnHmn-Hm -0.622 -0.302
## condtnAI-AI -0.638 -0.309  0.397
## srprs:cnH-H -0.251 -0.517  0.492  0.160
## srprs:AI-AI -0.301 -0.621  0.187  0.552  0.321

##
## Condition-specific novelty-to-resonance slopes:

##  condition   surprise.trend     SE   df lower.CL upper.CL
##  Human-AI             1.117 0.0205 2031    1.077    1.157
##  Human-Human          0.900 0.0339 2070    0.834    0.967
##  AI-AI                0.977 0.0259 2097    0.926    1.027
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```



8 Supplemental: Role-Level View



9 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot}=1} - \beta_{\text{surprise, slot}=2}|$ from `resonance ~ surprise * slot + (1|conversation_id)`. Per-story OLS slope differences are reported as a robustness layer.

Supplemental views retained:

- slot mean differences per metric (LMM `value ~ slot * condition + (1|story)`)
- condition-level marginal distributions and pooled `condition` LMMs
- pooled novelty-to-resonance model with `surprise * condition` interaction

- within-condition role box-plots (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type conditions)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)
- length-controlled variant (`+ nwords`) mirroring the paper

Analysis completed: 2026-04-23 12:22:37.793852